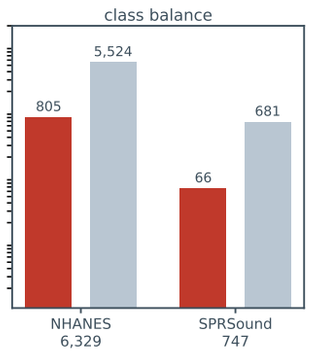
1. Data Acquisition

NHANES 2007–2012
clinical + spirometry

SPRSound (CC BY 4.0)
paediatric lung sound

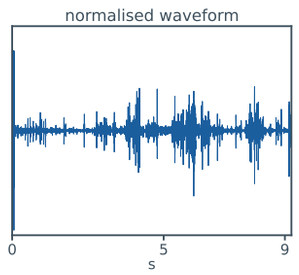


2. Preprocessing

exclude post-diagnosis vars
(MCQ040 / MCQ050)

deduplicate recordings
20,457 → 6,567

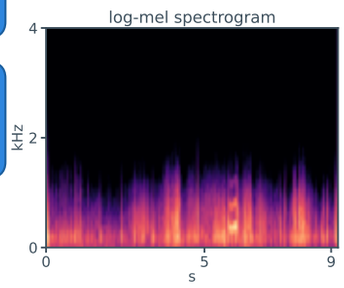
DC removal + peak
normalisation



3. Feature Extraction

14 symptom + 7 pulmonary
clinical features

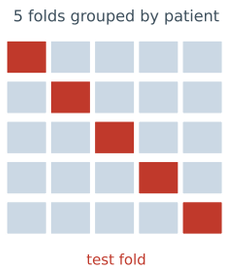
58 named acoustic
+ 768-d AST embedding



4. Data Splitting

stratified 5-fold CV

grouped by patient ID
(no cross-fold leakage)



5. Modality Agent Training

Symptom Agent
ROC-AUC 0.863

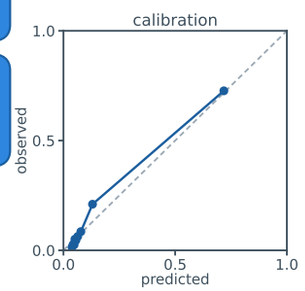
Pulmonary Agent
ROC-AUC 0.646

Acoustic Agent
ROC-AUC 0.732

6. Calibration & Evidence

sigmoid calibration
(train folds only)

evidence bundle:
prob · factors · flag



7. Decision Agent

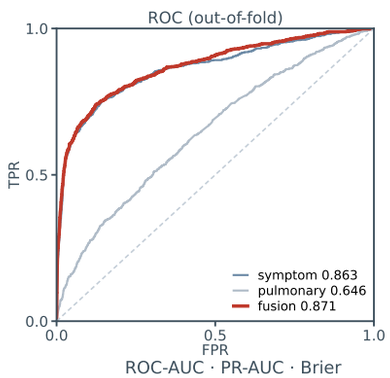
logistic meta-learner

abstain on low confidence

flag inter-agent conflict

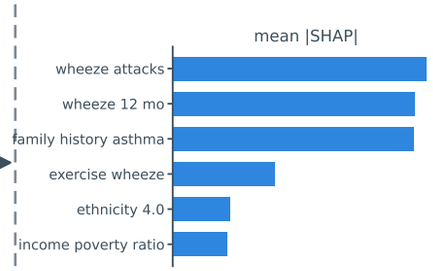
fused ROC-AUC 0.871

8. Evaluation



ROC-AUC · PR-AUC · Brier

9. Explanation Fusion



deletion-verified faithfulness

comprehensiveness 2.8× random

10. Clinical Output

Asthma likely

Asthma unlikely

Refer for review